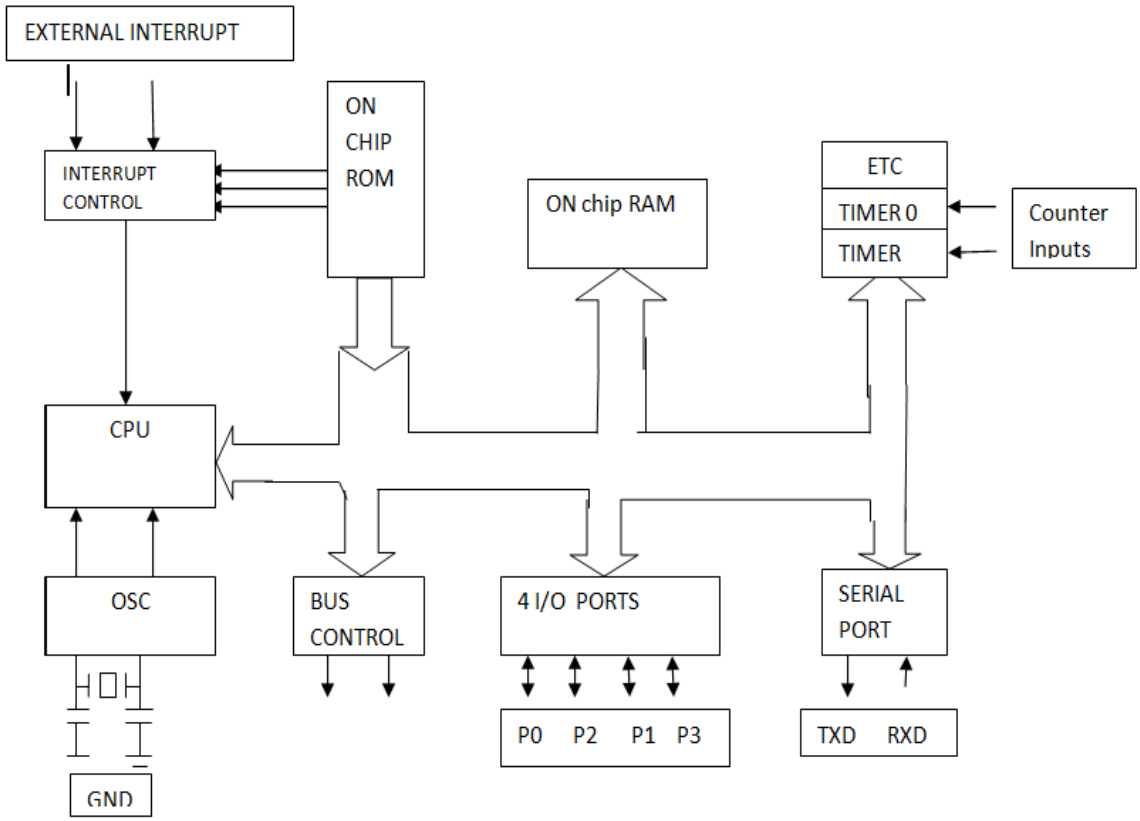
8051 Architecture ,Memory Organization And Special Functions Registers(SFR'S)

8051 Architecture

8051 microcontroller is designed by Intel in 1981. It is an 8-bit microcontroller. It is built with 40 pins DIP (dual inline package), 4kb of ROM storage and 128 bytes of RAM storage, 2 16-bit timers. It consists of four parallel 8-bit ports, which are programmable as well as addressable as per the requirement. An on-chip crystal oscillator is integrated in the microcontroller having crystal frequency of 12 MHz.

In the following diagram, the system bus connects all the support devices to the CPU. The system bus consists of an 8-bit data bus, a 16-bit address bus and bus control signals. All other devices like program memory, ports, data memory, serial interface, interrupt control, timers, and the CPU are all interfaced together through the system bus.



Basic components present internally inside 8051 Microcontroller architecture are:

CPU (Central Processing Unit): It synchronizes and manages all processes that are carried out in microcontroller. CPU contains ALU i.e. arithmetic and logic unit. This unit performs arithmetic and logic operations. Individual bit of register can be set, reset, cleared, complimented, tested, used in logical computation. It also contains registers used to store data temporarily.

Interrupts: Interrupts is a sub-routine call that given by the microcontroller when some other program with high priority is request for acquiring the system buses then interrupts occur in current running program.

Types of interrupt in 8051 Microcontroller:

- Timer 0 overflow interrupt - TF0
- Timer 1 overflow interrupt - TF1
- External hardware interrupt - INT0
- External hardware interrupt - INT1
- Serial communication interrupt - RI/TI

Memory: For operation Micro-controller required a program. This program guides the microcontroller to perform the specific tasks. This program installed in microcontroller required some on chip memory for the storage of the program.

Microcontroller also required memory for storage of data and operands for the short duration. In microcontroller 8051 there is code or program memory of 4 KB that is it has 4 KB ROM and it also comprise of data memory (RAM) of 128 bytes.

Bus : Bus is a group of wires which uses as a communication canal or acts as means of data transfer. The different bus configuration includes 8, 16 or more cables. Therefore, a bus can bear 8 bits, 16 bits all together.

Types of buses in 8051 Microcontroller:

Let's see the two types of bus used in 8051 microcontroller:

- **Address Bus**: 8051 microcontrollers is consisting of 16 bit address bus. It is used to address memory locations and to transfer the address from CPU to Memory of the microcontroller
- **Data bus**: 8051 microcontroller is consisting of 8 bits data bus. It is generally be used for transferring the data from one peripherals position to other peripherals.

Oscillator: As the microcontroller is digital circuit therefore it needs timer for their operation. To perform timer operation inside microcontroller it required externally connected or on-chip oscillator. Microcontroller is used inside an embedded system for managing the function of devices. Therefore, 8051 uses the two 16 bit counters and timers. For the operation of this timers and counters the oscillator is used inside microcontroller.

Input/Output Port:

Normally microcontroller is used in embedded systems to control the operation of machines in the microcontroller. Therefore, to connect it to other machines, devices or peripherals we require I/O interfacing ports in the microcontroller interface. For this purpose microcontroller 8051 has 4 input, output ports to connect it to the other peripherals.

Timers/Counters: 8051 microcontroller has two 16 bit timers and counters. These counters are again divided into a 8 bit register. The timers are used for measurement of intervals to determine the pulse width of pulses.

Registers:

Registers are used in the CPU to store information on temporarily basis which could be data to be processed, or an address pointing to the data which is to be fetched. In 8051, there is one data type is of 8-bits, from the MSB (most significant bit) D7 to the LSB (least significant bit) D0. With 8-bit data type, any data type larger than 8-bits must be broken into 8-bit chunks before it is processed.

Registers in 8051

Accumulator

The accumulator, register A, is used for all arithmetic and logic operations. If the accumulator is not present, then every result of each calculation (addition, multiplication, shift, etc.) is to be stored into the main memory. Access to main memory is slower than access to a register like the accumulator because the technology used for the large main memory is slower (but cheaper) than that used for a register.

The "R" Registers

The "R" registers are a set of eight registers, namely, R0, R1 to R7. These registers function as auxiliary or temporary storage registers in many operations.

The "B" Register

The "B" register is very similar to the Accumulator in the sense that it may hold an 8-bit (1-byte) value. The "B" register is used only by two 8051 instructions: **MUL AB** and **DIV AB**. To quickly and easily multiply or divide A by another number, you may store the other number in "B" and make use of these two instructions.

The Data Pointer

The Data Pointer (DPTR) is the 8051's only user-accessible 16-bit (2-byte) register. The Accumulator, R0–R7 registers and B register are 1-byte value registers. DPTR is meant for pointing to data. It is used by the 8051 to access external memory using the address indicated by DPTR. DPTR is the only 16-bit register available and is often used to store 2-byte values.

The Program Counter

The Program Counter (PC) is a 2-byte address which tells the 8051 where the next instruction to execute can be found in the memory. PC starts at 0000h when the 8051 initializes and is incremented every time after an instruction is executed. PC is not always incremented by 1. Some instructions may require 2 or 3 bytes; in such cases, the PC will be incremented by 2 or 3. **Branch, jump**, and **interrupt** operations load the Program Counter with an address other than the next sequential location.

The Stack Pointer (SP)

The Stack Pointer may hold an 8-bit (1-byte) value. The Stack Pointer tells the location from where the next value is to be retrieved from the stack. When a value is pushed onto the stack, the value of SP is incremented and then the value is stored at the resulting memory location. When a value is popped off the stack, the value is returned from the memory location indicated by SP, and then the value of SP is decremented.

This order of operation is important. SP will be initialized to 07h when the 8051 is initialized. If a value is pushed onto the stack at the same time, the value will be stored in the internal RAM address 08h because the 8051 will first increment the value of SP (from 07h to 08h) and then will store the pushed value at that memory address (08h). SP is modified directly by the 8051 by six instructions: PUSH, POP, ACALL, LCALL, RET, and RETI.

8051 Flag Bits and PSW Register

The program status word (PSW) register is an 8-bit register, also known as **flag register**. It is of 8-bit wide but only 6-bit of it is used. The two unused bits are **user-defined flags**. Four of the flags are called **conditional flags**, which means that they indicate a condition which results after an instruction is executed. These four are **CY** (Carry), **AC** (auxiliary carry), **P** (parity), and **OV** (overflow). The bits RS0 and RS1 are used to change the bank registers. The following figure shows the program status word register.

The PSW Register contains that status bits that reflect the current status of the CPU.



CY	PSW.7	Carry Flag
AC	PSW.6	Auxiliary Carry Flag
F0	PSW.5	Flag 0 available to user for general purpose.
RS1	PSW.4	Register Bank selector bit 1
RS0	PSW.3	Register Bank selector bit 0
OV	PSW.2	Overflow Flag
-	PSW.1	User definable FLAG
P	PSW.0	Parity FLAG. Set/ cleared by hardware during instruction cycle to indicate even/odd number of 1 bit in accumulator.

We can select register bank using RS0 and RS1

RS1	RS0	register bank	Address
0	0	0	00H- 07H
0	1	1	08H-0FH
1	0	2	10H-17H
1	1	3	18H-1FH

CY, the carry flag − This carry flag is set (1) whenever there is a carry out from the D7 bit. It is affected after an 8-bit addition or subtraction operation. It can also be reset to 1 or 0 directly by an instruction such as "SETB C" and "CLR C" where "SETB" stands for set bit carry and "CLR" stands for clear carry.

- **AC, auxiliary carry flag** − If there is a carry from D3 and D4 during an ADD or SUB operation, the AC bit is set; otherwise, it is cleared. It is used for the instruction to perform binary coded decimal arithmetic.

- **P, the parity flag** − The parity flag represents the number of 1's in the accumulator register only. If the A register contains odd number of 1's, then P = 1; and for even number of 1's, P = 0.

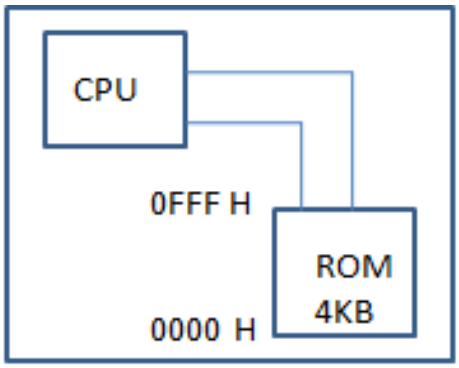
- **OV, the overflow flag** − This flag is set whenever the result of a signed number operation is too large causing the high-order bit to overflow into the sign bit. It is used only to detect errors in signed arithmetic operations.

8051 Memory Organization

The 8051 microcontroller has 128 bytes of Internal RAM and 4kB of on chip ROM.The RAM is also known as Data memory and the ROM is known as program memory.he program memory is also known as Code memory .This Code memory holds the actual 8051 program that is to be executed. In 8051 this memory is limited to 64K .Code memory may be found on-chip, as ROM or EPROM. It may also be stored completely off-chip in an external ROM or, more commonly, an external EPROM. The 8051 has only 128 bytes of Internal RAM but it supports 64kB of external RAM. As the name suggests, external RAM is any random access memory which is off-chip. Since the memory is off-chip it is not as flexible in terms of accessing, and is also slower.

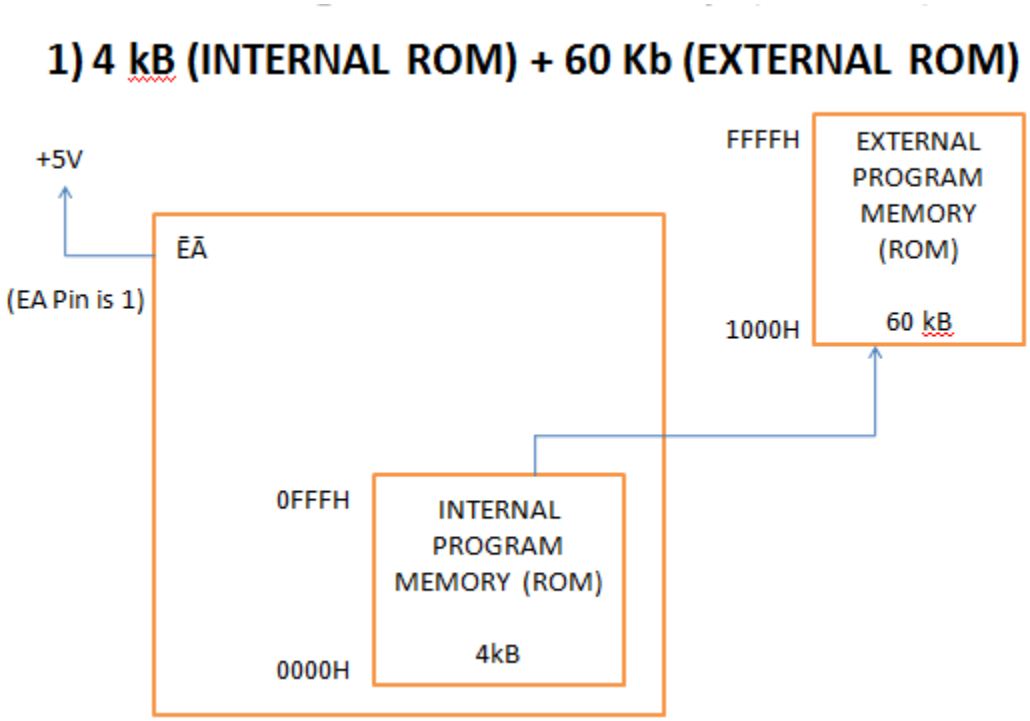
Program Memory (ROM) of 8051 Microcontroller

In 8051 Microcontroller, the code or instructions to be executed are stored in the Program Memory, which is also called as the ROM of the Microcontroller. The original 8051 Microcontroller by Intel has 4KB of internal ROM.



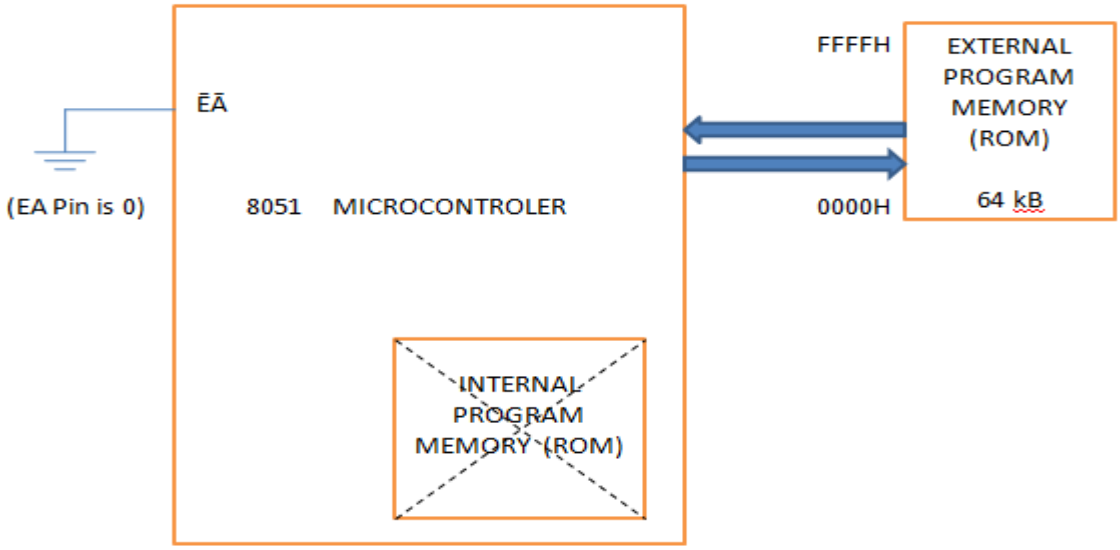
In case of 4KB of Internal ROM, the address space is 0000H to 0FFFH. If the address space i.e., the program addresses exceed this value, then the CPU will automatically fetch the code from the external Program Memory.

Program memory accessed through EA pin. In program memory two categories takes place. In 8051, the PSEN = 1 (is active) when reading a byte from external program memory (ROM). The command used to access external ROM is,



If EA is high, internal program memory is accessed to 0FFFH memory location and external program memory accessed from 1000H to FFFFH memory locations.

2) USE ONLY 64 Kb (EXTERNAL ROM)



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If EA is low, only external program memory accessed from 0000H to FFFFH memory locations.

Internal RAM OF 8051 : Layout of the RAM in 8051

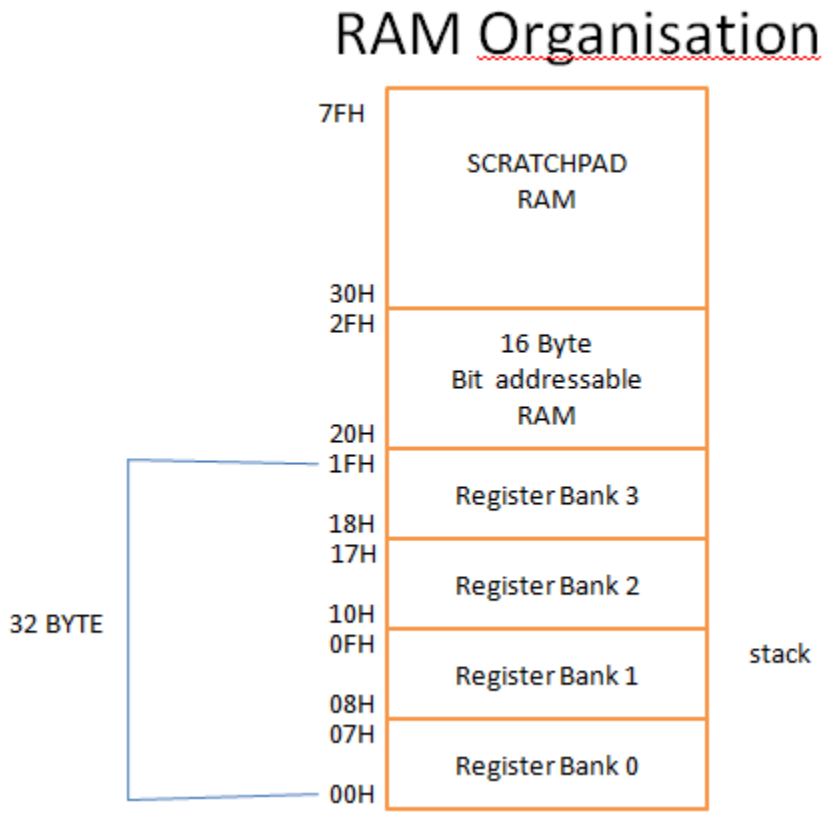
The data memory in 8051 is divided into three parts:

1. Lower 128 bytes (00H – 7FH), which are addressed by either Direct or Indirect addressing. Further, the Lower 128 bytes are divided into three parts,
    1. Register Banks (Bank 0,1,2,3) from 00H to 1FH – 32 bytes
    2. Bit Addressable Area from 20H to 2FH – 16 bytes
    3. General Purpose Register (Scratch Pad Area) from 30H to 7FH – 80 bytes

2. Upper 128 bytes (80H – 0FFH) for the Special Function Register (SFRs) which includes I/O ports (P0, P1, P2, P3), Accumulator (A), Timers(THx, TLx, TMOD, TCON, PCON), Interrupts(IE, IP), Serial Communication controls(SBUF, SCON), Program Status Word (PSW). These are addressed using Direct addressing.

3. 128 bytes of Additional Memory

Lower 128 bytes are as shown below.

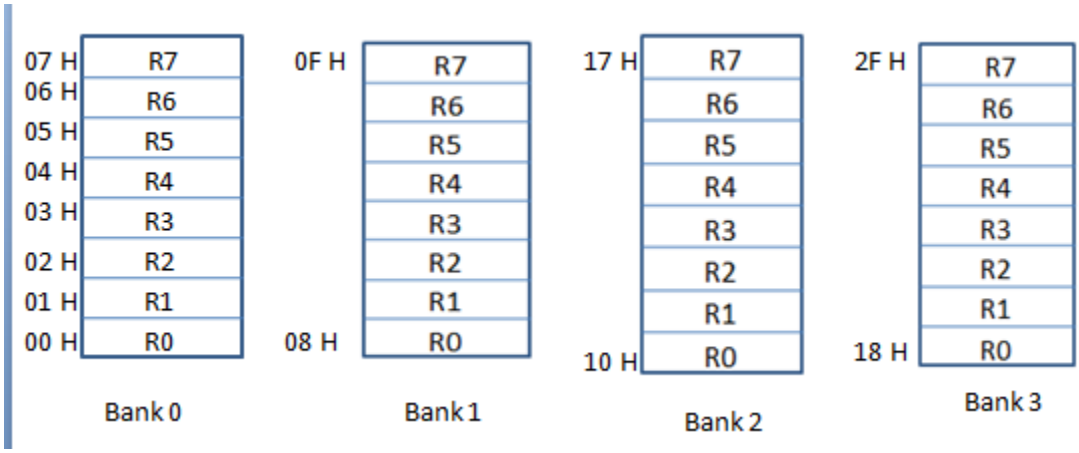


Register Banks

Registers are used to store information temporarily. But dividing into register banks is an interesting concept that helps utilize memory effectively. It allows you to address an entire range of memory using just the register's name.

The 8051 has four Register Banks with each bank having 8 "R" registers, which are used in many of its instructions. These R registers are numbered from 0 through 7 (R0, R1, R2, R3, R4, R5, R6, and R7).

These registers are generally used to assist in manipulating values and moving data from one memory location to another.



Bit addressable memory area

16 bytes of RAM is allotted as a bit addressable memory. It consists of 16 x 8 = 128 bits. These 128 bits can be individually bit-addressed, starting from 00H to 7FH or can be byte-addressable for 20H to 2FH. These bits can be modified by using SETB and CLR commands.

Bit addressing is very useful when we want to monitor the status of a particular pin of any port (say, P1.1) i.e., if we want to turn ON/OFF an LED connected to a particular pin we don't need to write the contents to the whole port, instead, we can address the particular pin to which the LED is connected and save memory.

General-purpose area/Scratchpad area

The rest of the RAM from 30H to 7FH, these 32 bytes can be addressed directly or indirectly using direct or indirect addressing modes, respectively. Therefore it's also called Scratch Pad Memory. You may read or write a full byte (8-bit) data at these locations.

SPECIAL FUNCTION REGISTERS(SFR):

To monitors or control the various functions within a microcontroller like control different hardware peripherals like Timers, Serial Port, I/O Ports etc. Here we use some special registers in 8051 microcontroller called ***Special Function Register (SRF)***

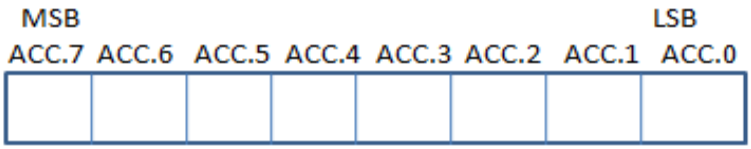
If we want to assess or change the operating mode of the 8051 Microcontroller. This will be done by manipulating the 8051 Microcontroller **Special Function Registers (SFRs).**

In 8051 micro controller there are **21 Special function registers (SFR)** and this includes I/O ports (P0, P1, P2, P3), Accumulator (A), Timers(THx, TLx, TMOD, TCON, PCON), Interrupts(IE, IP), Serial Communication controls(SBUF, SCON), Program Status Word (PSW). These are addressed using Direct addressing So, it required 21 unique locations for these 21 special function registers and the size of each register is of 1 byte. Some of these special function registers are **bit addressable registers** (which means you can access 8 individual bits inside a single byte), while some others are **only byte addressable register**.

F8H									FFH
F0 H	B								F7 H
E8 H									EF H
E0 H	ACC								E 7H
D8 H									DFH
D0 H	PSW								D7 H
C8 H									CF H
C0 H									C7H
B8H	IP								BF H
B0 H	P3								B7 H
A8 H	IE								AFH
A0 H	P2								A7 H
98 H	SCON	SBUF							9FH
90 H	P1								97 H
88 H	TCON	TMOD	TL0	TL1	TH0	TH1			8FH
80 H	P0	SP	DPL	DPH				PCON	87 H

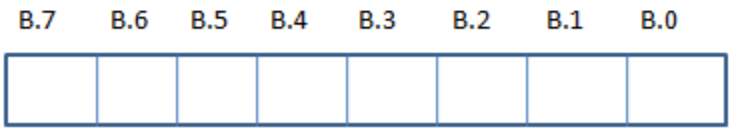
Accumulator(A Register)

- 8 bit register
- Holds data and receive result of arithmetic instructions
- Bit and byte addressable
- Accessed by direct addressing
- Physical address E0 H



B Register

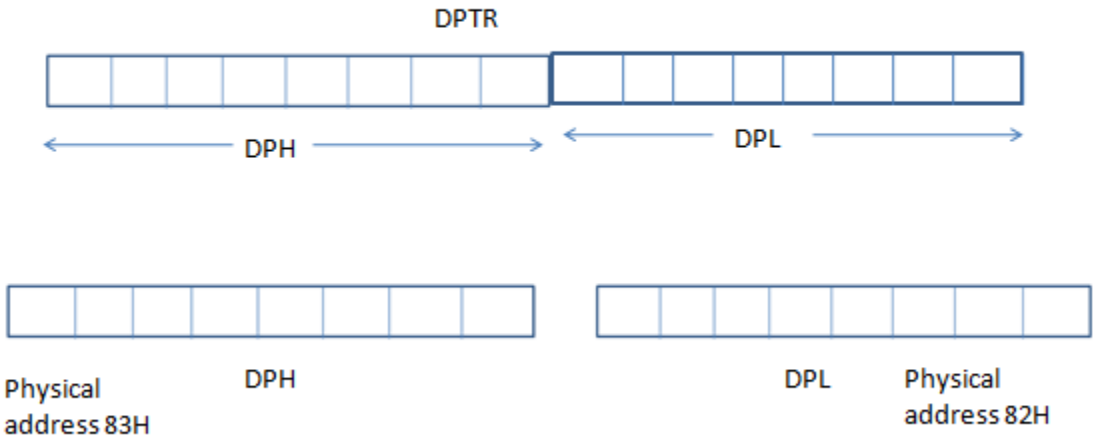
- 8 bit register
- Bit and byte addressable
- Only used for multiplication and division operation
- Physical address F0 H



DPTR(DATA POINTER REGISTER)

- 16 Bit register used to hold 16 bit address of data memory.
- Can be used as 2 numbers of 8 bit data pointer DPH and DPL.
- 8 bit data pointers are used for accessing internal RAM and SFR.
- 16 bit data pointer is used for accessing external memory.

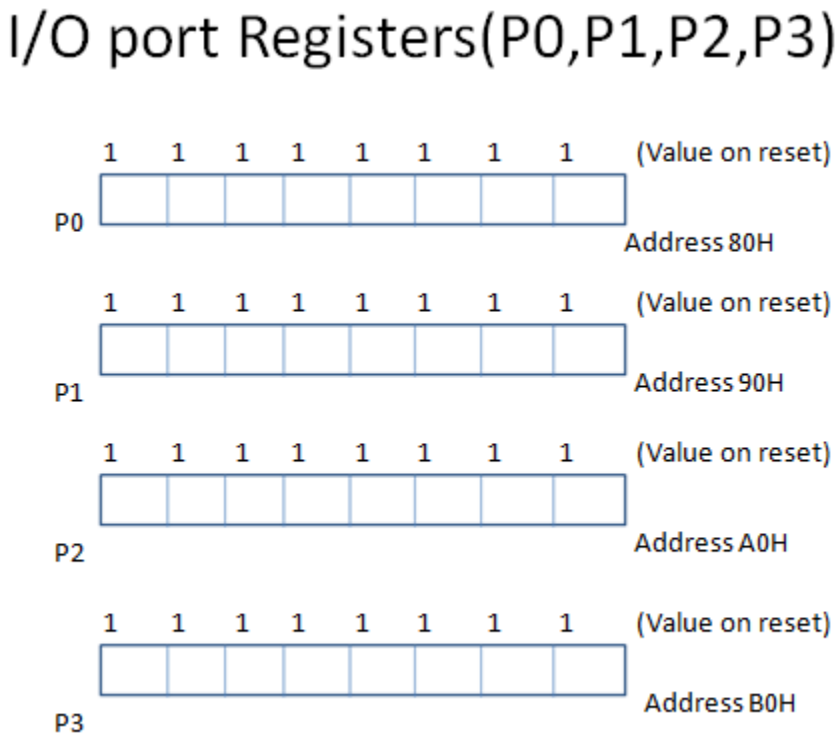
DPTR(Data pointer Register)



I/O Port Registers

8051 microcontrollers have 4 bidirectional I/O ports. Port 0, Port 1, Port 2 and Port 3 (P0, P1, P2 and P3). Which can be work as input or output port. Each port having 8-bits. Hence, total 32 input/output pins allow

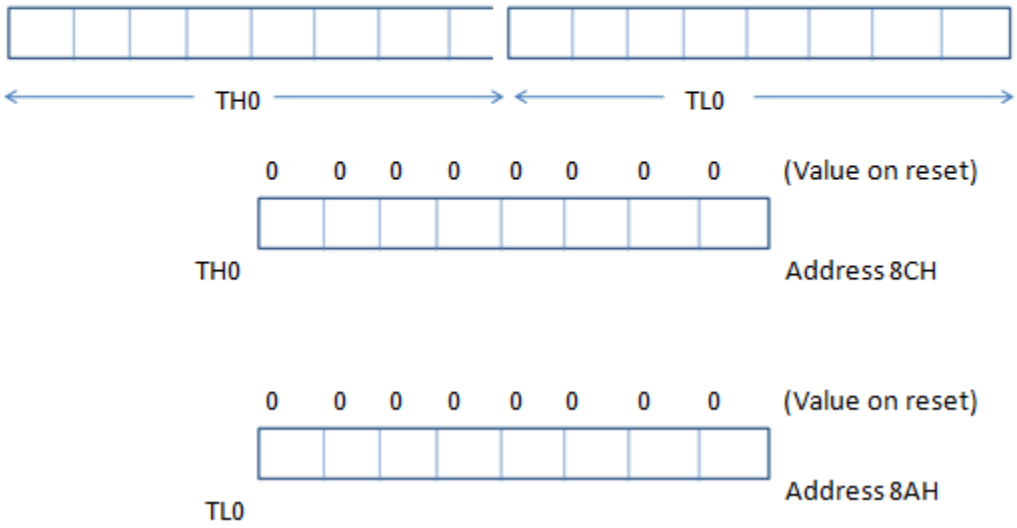
the microcontroller to communicate with outside world means this port allow to be connected with the peripheral devices.



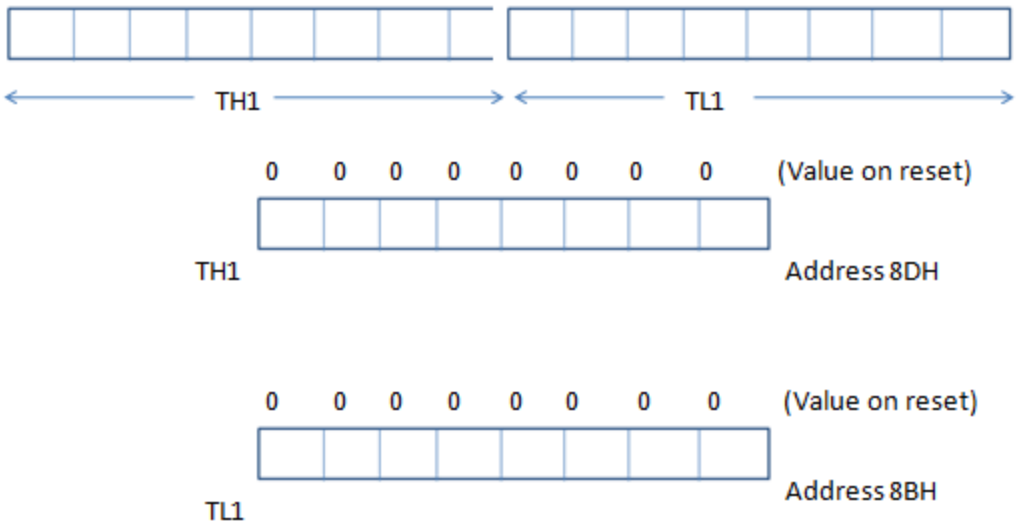
Timers and Counters

- 2 16 bit programmable timers/counters(Timer 0,Timer 1)
- Can be used as a timer to generate time delay or as a counter for counting.
- TMOD(Timer Mode Register) and TCON(Timer Control Register).Used for activating and configuring timers and counters.

Timer 0 Register



Timer 1 Register



TMOD(Timer Mode) Register

- 8 bit register

D7	D6	D5	D4	D3	D2	D1	D0
GATE	C/T	M1	M0	GATE	C1/T	M1	M0
Timer1/C1				**Timer0/C0**			

Address 89H

Bit	Symbol	Description
7 / 3	Gate	OR Gate Enable Bit
6 / 2	C/Tx	Select Timer or Counter mode
5 / 1	TxM1	Timer / Counter Operating Mode Select Bit 1
4 / 0	TxM0	Timer / Counter Operating Mode Select Bit 0

The Gatex bit is used to operate the Timerx with respect to the INTx pin or regardless of the INTx pin.

- GATE1 = 1 ==> Timer1 is operated only if TR1 (in TCON) is SET and Signal on INT1 is HIGH.
- GATE1 = 0 ==> Timer1 is operated irrespective of Signal on INT1 pin but TR1 (in TCON) must be SET.
- GATE0 = 1 ==> Timer0 is operated only if TR0 (in TCON) is SET and Signal on INT0 is HIGH.
- GATE0 = 0 ==> Timer0 is operated irrespective of Signal on INT0 pin but TR0 (in TCON) must be SET.

The C/Tx bit is used selects the source of pulses for the Timer to count.

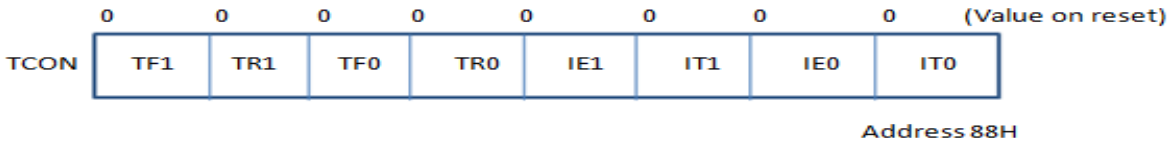
- C/T1 = 1 ==> Timer1 counts pulses from Pin T1 (P3.5) (Counter Mode)
- C/T1 = 0 ==> Timer1 counts pulses from internal oscillator (Timer Mode)
- C/T0 = 1 ==> Timer0 counts pulses from Pin T0 (P3.4) (Counter Mode)
- C/T0 = 0 ==> Timer0 counts pulses from internal oscillator (Timer Mode)

- **Mode selects bits of TMODE register:** The M1 and M0 are mode select bits, which are used to select the timer operations. There are four modes to operate the timers.

M1	M0	Mode of operation
0	0	Mode 0 (13-bit timer mode)
0	1	Mode 1 (16-bit timer mode)
1	0	Mode 2 (8-bit Auto Reload mode)
1	1	Mode 3 (Split timer mode)

Mode Selection Bits Of TMODE Register

TCON(Timer Control)Register



Timer Control or TCON Register is used to start or stop the Timers of 8051 Microcontroller. It also contains bits to indicate if the Timers has overflowed. The TCON SFR also consists of Interrupt related bits.

Bit	Symbol	Description	Additional Info
7	TF1	Timer 1 Overflow Flag	Set when Timer 1 overflows (all 1s to 0). Cleared when processor executes ISR at 001BH.
6	TR1	Timer 1 Run Control Bit	To enable Timer/Counter, set to 1. Clear to halt the timer.
5	TF0	Timer 0 Overflow Flag	Set when Timer 0 overflows (all 1s to 0). Cleared when

			processor executes ISR at 000BH.
4	TR0	Timer 0 Run Control Bit	To enable Timer/Counter, set to 1. Clear to halt the timer.
3	IE1	Ext. Interrupt 1 Edge Flag	Set when HIGH to LOW is received on INT1 (P3.3). Cleared when processor executes ISR at 0013H.
2	IT1	Ext. Interrupt 1 Type Control Bit	If 1, Interrupt 1 occurs on falling edge. If 0, Interrupt 1 occurs on low level.
1	IE0	Ext. Interrupt 0 Edge Flag	Set to 1 when HIGH to LOW is received on INT0 (P3.2). Cleared when processor executes ISR at 0003H.
0	IT0	Ext. Interrupt 0 Type Control Bit	If 1, Interrupt 0 occurs on falling edge. If 0, Interrupt 0 occurs on low leve

Peripheral Control Registers

PCON (Power Control)

The PCON or Power Control register, as the name suggests is used to control the 8051 Microcontroller's Power Modes and is located at 87H of the SFR Memory Space. Using two bits in the PCON Register, the microcontroller can be set to Idle Mode and Power Down Mode.

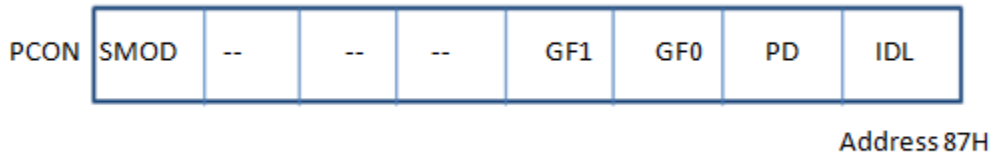
NOTE: PCON register is not bit addressable.

During Idle Mode, the Microcontroller will stop the Clock Signal to the ALU (CPU) but it is given to other peripherals like Timer, Serial, Interrupts, etc. In order to terminate the Idle Mode, you have to use an Interrupt or Hardware Reset.

In the Power Down Mode, the oscillator will be stopped and the power will be reduced to 2V. To terminate the Power Down Mode, you have to use the Hardware Reset.

Apart from these two, the PCON Register can also be used for few additional purposes. The SMOD Bit in the PCON Register is used to control the Baud Rate of the Serial Port.

There are two general purpose Flag Bits in the PCON Register, which can be used by the programmer during execution.



The following table describes the function of each bit in the PCON Register.

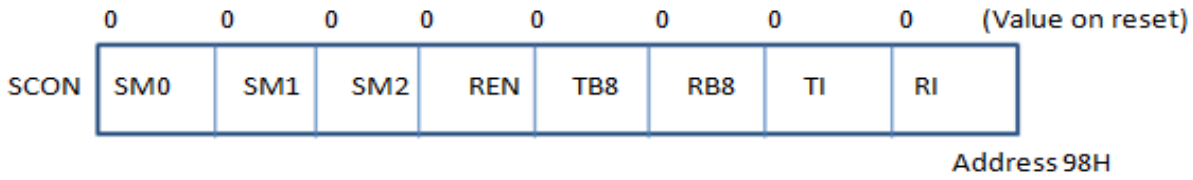
Bit	Symbol	Description	Additional Info
7	SMOD	Serial communication baud Rate Modify bit	If 1, doubles the baud rate using Timer 1. If 0, normal timer 1 baud rate.
6 – 4	___	—	
3	GF1	General Purpose User Flag (Bit 1)	

2	GF0	General Purpose User Flag (Bit 0)	
1	PD	Power Down Bit	To enter Power Down Mode, set to 1
0	IDL	Idle Mode Bit	To enter Idle Down Mode, set to 1

SCON (Serial Control)

The Serial Control or SCON SFR is used to control the 8051 Microcontroller's Serial Port. It is located as an address of 98H. Using SCON, you can control the Operation Modes of the Serial Port, Baud Rate of the Serial Port and Send or Receive Data using Serial Port.

SCON Register also consists of bits that are automatically SET when a byte of data is transmitted or received.



Bit	Symbol	Description
7	SM0	Serial Port Mode Selection Bit 0
6	SM1	Serial Port Mode Selection Bit 1
5	SM2	Multiprocessor Comm. Bit
4	REN	Set /Cleared by software to enable or disable reception
3	TB8	The 9th bit transmitted in modes 2 and 3
2	RB8	The 9 th data bit that is received in modes 2 and 3
1	TI	Transmit Interrupt Flag (set by hardware)
0	RI	Receive Interrupt Flag(set by hardware)

The Serial Port Mode Selection Bits (SM0 and SM1) determine the mode of UART and also the baud rate.

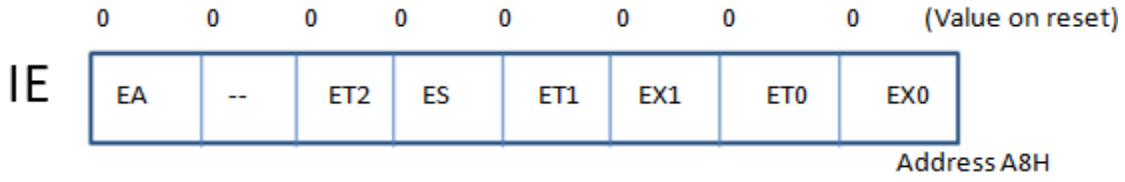
Serial Port Mode Selection Bits

SM0	SM1	Mode	Description	Baud Rate
0	0	0	8-Bit Synchronous Shift Register Mode	Fixed Baud Rate(Frequency of oscillator / 12)

0	1	1	8-bit Standard UART mode	Variable Baud Rate (Can be set by Timer 1)
1	0	2	9-bit UART	Fixed Baud Rate (Frequency of oscillator / 32 or Frequency of oscillator / 64
1	1	3	9-bit UART	Variable Baud Rate (Can be set by Timer 1)

IE (Interrupt Enable)

The IE or Interrupt Enable Register is used to enable or disable individual interrupts. If a bit is SET, the corresponding interrupt is enabled and if the bit is cleared, the interrupt is disabled. The Bit7 of the IE register i.e., EA bit is used to enable or disable all the interrupts.

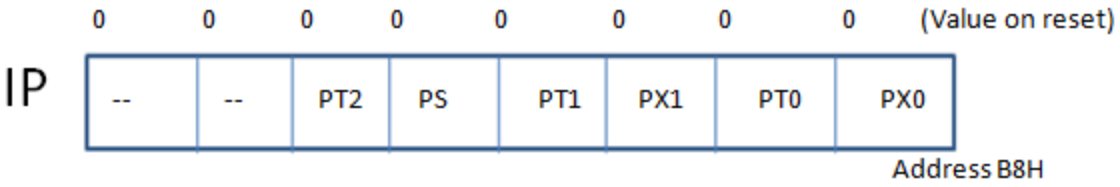


Bit	Symbol	Description	Additional Info
7	EA	Global Interrupt Enable Bit.	If set to 1, individual interrupts can be enabled or disabled by setting or clearing its enable bit. If set to 0, all interrupts are disabled.

6	—	—	
5	ET2	Reserved	
4	ES	Serial Port Interrupt Enable Bit	If set to 1, Serial Port interrupt is enabled. If set to 0, Serial Port interrupt is disabled.
3	ET1	Timer 1 Overflow Interrupt Enable Bit	If set to 1, Timer 1 Overflow interrupt is enabled. If set to 0, Timer 1 Overflow interrupt is disabled.
2	EX1	Ext. Interrupt 1 Enable Bit	If set to 1, Ext. Interrupt 1 is enabled. If set to 0, Ext. Interrupt 1 is disabled.
1	ET0	Timer 0 Overflow Interrupt Enable Bit	If set to 1, Timer 0 Overflow interrupt is enabled. If set to 0, Timer 0 Overflow interrupt is disabled.
0	EX0	Ext. Interrupt 0 Enable Bit	If set to 1, Ext. Interrupt 0 is enabled. If set to 0, Ext. Interrupt 0 is disabled.

IP (Interrupt Priority)

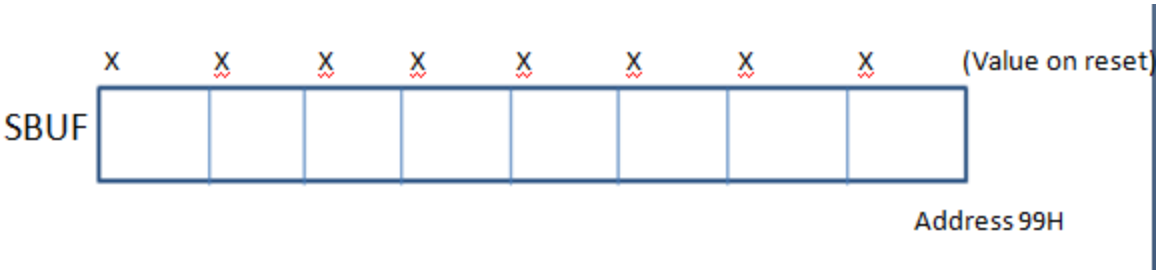
The IP or Interrupt Priority Register is used to set the priority of the interrupt as High or Low. If a bit is CLEARED, the corresponding interrupt is assigned low priority and if the bit is SET, the interrupt is assigned high priority.



Bit	Symbol	Description
7	—	—
6	—	—
5	PT2	Reserved
4	PS	Priority of Serial Port Interrupt
3	PT1	Timer 1 Interrupt priority level
2	PX1	Priority of Ext. Interrupt 1
1	PT0	Timer 0 Interrupt priority level
0	PX0	Priority of Ext. Interrupt 0

SBUF (Serial Data Buffer)

The Serial Buffer or SBUF register is used to hold the serial data while transmission or reception.


X X X X X X X X
(Value on reset)
SBUF
Address 99H